

HRISHABH SHAH

Software Engineer • DevOps & Platform Engineering

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Summary

Software Engineer at Siemens specializing in backend systems and platform engineering. Build and operate an internal developer platform (SCORE) on Kubernetes and AWS — a Backstage developer portal, Go control-plane controllers (UCP), and GitOps delivery templates that promote workloads from dev to QA to production, with a focus on automation, reliability, and observability.

Technical Skills

Languages	C/C++, JavaScript, Python, Java, TypeScript, SQL
Backend / Web	Node.js, Backstage, Spring Boot, React.js, Angular, REST APIs, Zod
Cloud / K8s	AWS (EKS, IAM, RDS, S3, ACM, CloudWatch), Kubernetes, Helm, Kustomize, EKS controllers (kubebuilder, controller-runtime)
GitOps / CI/CD	ArgoCD, FluxCD, GitHub Actions, GitLab CI, KubeVela, Docker, Trivy
Observability	Grafana, Prometheus, Datadog, AWS CloudWatch (metrics, logs, dashboards, alerting)
Automation / IaC	Chef (cookbooks, recipes, scripting), Ansible, Terraform-style ACK/CRDs, Linux CLI, Bash
Core	DSA, OOP, System Design, Database Design, Testing (envtest, chainsaw, Vitest), SDLC

Experience

Software Engineer — Siemens (STSPL)

Aug 2025 – Present

Platform Engineering / Internal Developer Platform (SCORE)

Bengaluru / Kolkata, India

- **Backstage (backend):** developed and maintained backend plugins and scaffolder actions in TypeScript on Node.js within a Yarn-4 monorepo of ~40+ plugins; added Zod-validated inputs, typed error handling, and GitLab, GitHub Enterprise, and Entra ID group integrations.
- **UCP (Universal Control Plane):** built and extended custom Kubernetes controllers in Go (controller-runtime / kubebuilder) using a hierarchical enterprise → tenant control-plane model (each an EKS cluster); implemented sub-reconcilers, finalizers, CRDs, and an ACL framework with idempotent, eventually-consistent reconciliation.
- **Delivery templates (dev → QA → prod):** authored reusable scaffolder templates and a three-tier ArgoCD GitOps model (Kustomize bases + per-environment overlays) that promotes applications and infrastructure across dev, QA, and production consistently.
- Provisioned AWS infrastructure declaratively (EKS, IAM, RDS, S3, ACM) via ACK CRDs; packaged and shipped services with Helm charts and Docker base images.
- Instrumented services and clusters with **Prometheus, Grafana, Datadog, and CloudWatch** — dashboards, metrics, log pipelines, and alerting — improving reliability and mean-time-to-detect.
- Wrote controller tests with envtest and e2e tests with chainsaw; hardened CI/CD gates (type-checks, lint, coverage, Trivy image scanning).

Apprentice — Technical, Siemens (STSPL)

Aug 2024 – Aug 2025

Backend Engineering & Cloud-Native Foundations

India

- Engineer and operate **SCORE**, Siemens' internal developer platform, spanning a Backstage developer portal, Go-based Kubernetes control-plane controllers, and GitOps delivery — enabling self-service provisioning for thousands of engineers.
- Built backend solutions during technical training and contributed to the SiLC demonstration project; developed foundations in backend engineering, containerization, Kubernetes, GitOps, and CI/CD.

Backend Development Intern — Xopun Tech

Jul 2023 – Sep 2023

RESTful Services

India

- Built RESTful backend services with Spring Boot — CRUD operations, authentication, and DB integration.

Machine Learning Intern — NIELIT

Aug 2022 – Sep 2022

Applied ML / Data Analysis

India

- Worked with Python-based machine-learning algorithms and data-analysis techniques.

Key Projects

- Backstage Developer Portal (SCORE)**

TypeScript, Node.js, Backstage, Helm, ArgoCD

- Backend plugins and scaffolder templates that automate repo creation, catalog onboarding, and Entra ID group registration as self-service actions; deployed via Helm and ArgoCD.
- Universal Control Plane (UCP)**

Go, controller-runtime, kubebuilder, EKS, ACK

- Custom operators reconciling tenant control planes and AWS resources with finalizers, ACL-based access control, and idempotent reconciliation; validated with envtest and chainsaw e2e suites.
- GitOps Onboarding & Promotion Pipeline (dev → QA → prod)**

ArgoCD, Kustomize, Helm

- Three-tier ArgoCD model (governance projects, application CRs, infra CRs) with per-environment overlays that standardize how apps and infrastructure are promoted across environments.
- Cloud-Native Portfolio**

React, TypeScript, Docker, Kubernetes, Helm, ArgoCD, FluxCD

- Production-style portfolio with GitHub Actions CI/CD, Trivy image scanning, GHCR publishing, Kubernetes manifests, Helm packaging, and GitOps delivery. [\[GitHub\]](#)
- Attendance Management System**

Angular, Spring Boot, TypeScript, Java, PostgreSQL

- Full-stack attendance tracking with backend APIs, responsive UI, real-time updates, and reporting.
- Library Management System**

Spring Boot, Java, MySQL, Spring Security

- Backend for library operations with authentication, catalog management, secure CRUD APIs, and DB integration.

Education

B.Tech, Computer Science & Engineering

Assam Engineering College

2020 – 2024

Assam, India

Achievements

- Technical Head — Coding Club, Assam Engineering College.
- 5-star HackerRank (Problem Solving); 200+ problems solved on LeetCode.
- 2nd place — Algosmash, Royal Global University; 3rd place — Codewar, Assam Engineering College.
- GATE score 472 (Rank 3784); CEE Rank 192.